GNPS formula-disjoint — 17 methods (consumed development)
Recall@1 ladder (rows preset by R@1; cells show within-metric rank)

	R@1	R@2	R@3	R@5	R@10	R@20	MRR	Mean rank	Median rank	Macro AUROC	Macro AP (hist. AUPRC)	Micro AUROC	Micro AP (hist. AUPRC)
full23_v132	89.6%	95.8%	97.7%	99.0%	99.7%	100.0%	93.7%	1.23	1.00	94.7%	93.7%	95.4%	87.4%
weighted_spectral_entropy	88.3%	94.5%	97.1%	98.7%	99.7%	100.0%	92.8%	1.28	1.00	93.9%	92.8%	92.8%	78.1%
p2b_noise_v1_frozen	88.3%	95.0%	97.3%	99.0%	99.7%	100.0%	92.9%	1.26	1.00	94.0%	92.9%	92.6%	76.8%
p2b_sqrt_cosine	88.2%	94.7%	97.1%	98.8%	99.6%	100.0%	92.8%	1.27	1.00	93.7%	92.8%	92.4%	76.7%
p2b_unweighted_entropy	88.2%	94.8%	97.2%	98.8%	99.6%	100.0%	92.8%	1.27	1.00	93.8%	92.8%	92.4%	76.9%
entropy_raw_public	88.1%	94.8%	97.4%	98.9%	99.6%	100.0%	92.8%	1.26	1.00	94.1%	92.8%	92.0%	71.9%
noise_v1	88.1%	95.2%	97.2%	98.8%	99.7%	100.0%	92.8%	1.26	1.00	93.8%	92.8%	93.1%	76.9%
denoising_search_public	88.0%	94.7%	97.2%	98.9%	99.6%	100.0%	92.7%	1.27	1.00	94.0%	92.7%	92.2%	73.1%
p2b_official_frozen	88.0%	94.8%	97.3%	99.0%	99.6%	100.0%	92.8%	1.27	1.00	93.9%	92.8%	92.5%	75.9%
neutral_loss_sqrt_cosine	87.9%	94.8%	97.3%	98.9%	99.7%	100.0%	92.7%	1.27	1.00	93.8%	92.7%	92.3%	76.2%
spec2vec_gnps_public	87.4%	94.6%	97.1%	98.7%	99.6%	100.0%	92.4%	1.29	1.00	93.4%	92.4%	92.0%	75.7%
spec2vec_2026_retrained	87.2%	94.5%	97.0%	98.7%	99.6%	100.0%	92.3%	1.29	1.00	93.3%	92.3%	92.0%	75.4%
cosine_greedy	87.2%	94.8%	97.0%	98.7%	99.6%	100.0%	92.3%	1.29	1.00	93.4%	92.3%	92.2%	77.8%
modified_cosine	87.2%	94.8%	97.0%	98.7%	99.6%	100.0%	92.3%	1.29	1.00	93.4%	92.3%	92.2%	77.8%
official_dreams	86.8%	94.4%	97.1%	98.7%	99.6%	100.0%	92.0%	1.30	1.00	93.2%	92.0%	90.7%	68.0%
ms2deepscore_2x_public	81.5%	91.7%	95.9%	98.2%	99.6%	100.0%	88.7%	1.41	1.00	90.1%	88.7%	88.6%	72.1%
specembedding_official	79.3%	90.0%	94.4%	97.5%	99.3%	99.9%	87.1%	1.51	1.00	88.5%	87.1%	82.5%	54.1%

Rank (1 = best)
17
16
15
14
13
12
11
10
9
8
7
6
5
4
3
2
1